DEGREE OF AN EFFECTIVE CARTIER DIVISOR ON A PROPER SCHEME OF DIMENSION AT MOST ONE AND THE DEGREE SHIFT BY THAT DIVISOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Additivity of the Euler characteristic and the finite-support computation. Stacks Project, Varieties, Lemmas lemma-euler-characteristic-additive (tag 08AA) and lemma-chi-tensor-finite (tag 0AYT). These produce the degree formula from the exact sequence.
- Effective Cartier divisors and their invertible sheaves. Stacks Project, Divisors, Definition divisors-definition-invertible-sheaf-effective-Cartier-divisor (tag 01WX). Supplies the exact sequence defining O_X(D).

Source reference. Original reference 07ZU: sources/stacks-chow/tags.txt, source lines 10159–10159. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 06LH: sources/stacks-chow/tags.txt, source lines 8466–8466. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02KA: sources/stacks-chow/tags.txt, source lines 3365–3365. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01WX: sources/stacks-chow/tags.txt, source lines 2498–2498. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08AA: sources/stacks-chow/tags.txt, source lines 10525–10525. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AYT: sources/stacks-chow/tags.txt, source lines 13801–13801. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field, let X be a proper scheme of dimension ≤ 1 over k. Let D be an effective Cartier divisor on X. Then D is finite over $\mathrm{Spec}(k)$ of degree $\deg(D) = \dim_k \Gamma(D, \mathcal{O}_D)$. For a locally free sheaf $\mathcal{E}$ of rank n we have*

$$\deg(\mathcal{E}(D)) = n \deg(D) + \deg(\mathcal{E})$$

where $\mathcal{E}(D) = \mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{O}_X(D)$.

Proof. Since D is nowhere dense in X (Divisors, Lemma 07ZU) we see that $\dim(D) \leq 0$. Hence D is finite over k by Lemma 06LH. Since k is a field, the morphism $D \to \mathrm{Spec}(k)$ is finite locally free and hence has a degree (Morphisms, Definition 02KA), which is clearly equal to $\dim_k \Gamma(D, \mathcal{O}_D)$ as stated in the lemma. By Divisors, Definition 01WX there is a short exact sequence

$$0 \to \mathcal{O}_X \to \mathcal{O}_X(D) \to i_* i^* \mathcal{O}_X(D) \to 0$$

where $i : D \to X$ is the closed immersion. Tensoring with $\mathcal{E}$ we obtain a short exact sequence

$$0 \to \mathcal{E} \to \mathcal{E}(D) \to i_* i^* \mathcal{E}(D) \to 0$$

The equation of the lemma follows from additivity of the Euler characteristic (Lemma 08AA) and Lemma 0AYT. □